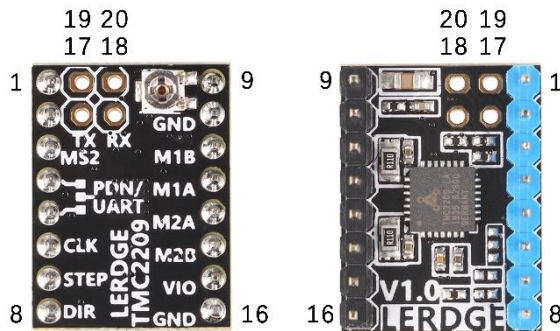


TMC2209 Lerdge-SilentStepDrive

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Module Top View



Features and additional Resources

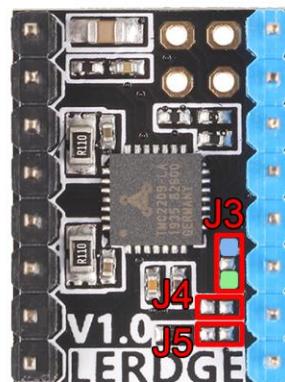
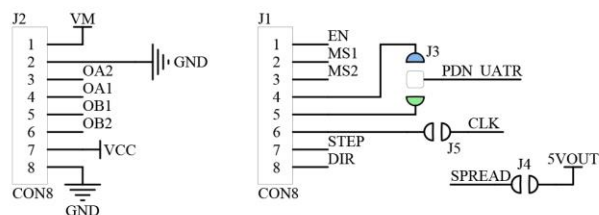
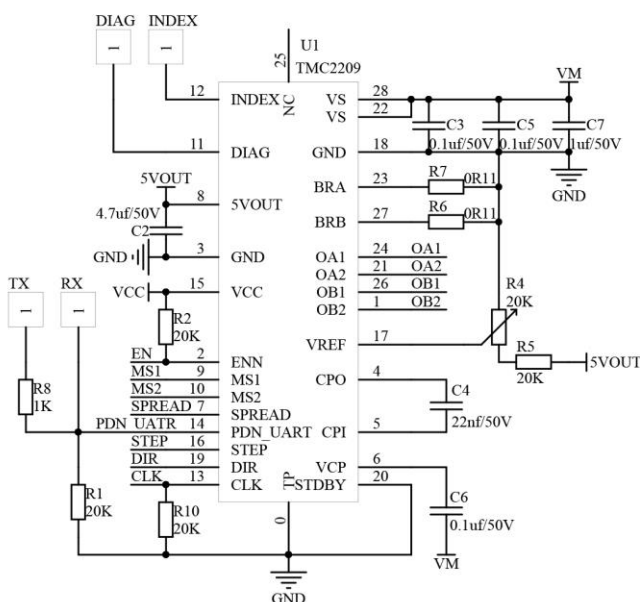
- ✓ TMC2209-LA stepper motor controller & driver
- ✓ Supply voltage 4.75-28V
- ✓ Continuous I_{phase} = 1.4ARMS
- ✓ I_{phase} up to 1.77ARMS = 2.5A_{peak} for a short time
- ✓ Quiet operation with StealthChop™
- ✓ Sensorless homing with StallGuard4™
- ✓ Energy savings with CoolStep™
- ✓ Configuration and extended diagnostics via UART
- ✓ Control via Step & Dir interface
- ✓ Board width 15.5mm, board height 20.5mm
- ✓ 2x8 pin 2.54mm header rows for pins/connectors

Pin List

19	DIAG	20	INDEX
17	TX	18	RX

1	EN	9	VM
2	MS1	10	GND
3	MS2	11	M1B
4	NC PDN_UART	12	M1A
5	NC PDN_UART	13	M2A
6	NC CLK	14	M2B
7	STEP	15	VIO
8	DIR	16	GND

Schematics



$$I_{RMS} = \frac{325mV}{R_{sense} + 20m\Omega} \cdot \frac{1}{\sqrt{2}} \cdot \frac{V_{ref}}{2.5V} \approx 0.88 \cdot V_{ref}$$

V_{ref} 0...2.5V (default: 1.25V)
R_{sense} 0.11 Ohm

EN (with pull-up)
GND Enable driver
VCC Disable driver

SPREAD (with pull-down)
GND stealthChop
VCC spreadCycle

MS1	MS2	Steps	Interpolation
GND	GND	8	256
VCC	VCC	16	256
VCC	GND	32	256
GND	VCC	64	256